

Weather Prediction Using Linear Regression

This project analyzes a **Seattle weather dataset** and builds a machine learning model to predict weather-related values, focusing on **maximum daily temperature** using Linear Regression.

Dataset Features

- **date** — Observation date
- **precipitation** — Rainfall amount
- **temp_max** — Maximum daily temperature
- **temp_min** — Minimum daily temperature
- **wind** — Wind speed
- **weather** — Condition (sun, rain, drizzle, fog, snow)

Tools & Libraries

- Python
- Pandas & NumPy
- Matplotlib
- Scikit-learn

Methodology

01

Load & Preprocess

Load dataset with Pandas; check data types, missing values, and duplicates.

03

Feature Selection

Inputs: **precipitation**, **temp_min**, **wind**. Target: **temp_max**.

02

Exploratory Data Analysis

Summary statistics, histograms, boxplots, and correlation analysis.

04

Train & Evaluate

Split data, train Linear Regression (Scikit-learn), evaluate with **MAE**, **MSE**, **RMSE**, and **R² Score**. Visualize Actual vs. Predicted and Residual plots.

Results & Conclusion

The trained model estimates maximum daily temperature from selected weather features. Evaluation metrics measure prediction accuracy while visualizations reveal model performance and error distribution. This project demonstrates a **complete ML workflow** — from data loading and exploration to training and evaluation — highlighting the importance of preprocessing, feature selection, and performance assessment. While Linear Regression suits continuous predictions like temperature, **classification algorithms** would better predict categorical conditions like rain or sunshine.